

Hey, I'm David.

# I void warranties.

Engineer and leader drawn to complex systems with a bias toward **adversarial thinking, incident response, resiliency, reliability, security, and privacy**. Google SRE, Deputy CIO of a 700-person org, and built both AI programs and platforms. Nights and weekends find me with a soldering iron, a software-defined radio, and blowing through AI usage limits.

- Go
- Python
- Kubernetes
- AI
- SDR
- Hardware
- Policy
- TS/SCI

## Experience & selected work

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| 2025 – now | <div>Consulting / AI &amp; software-defined radio</div> <div>INDEPENDENT</div> <p>Independent engagements across <b>applied AI and software-defined radio</b> — public-safety comms tracking, SDR orchestration on Kubernetes, and agentic-AI workflows — plus technical expert-witness work. Recent builds:</p> <ul style="list-style-type: none"><li>• <b>trunk'd</b> — Public-safety radio, made legible — live SDR capture, AI transcription and incident board. RF to browser in under a second.</li><li>• <b>Vega-Bray Observatory</b> — A 20" mount, dark for 15 years, reverse-engineered into ESP32 firmware and a Go pointing service.</li><li>• <b>sdrlight</b> — An in-progress attempt to schedule commodity radios like compute — a zero-copy SDR mesh on unprivileged Kubernetes pods.</li><li>• <b>Luna</b> — A 1-meter lunar relief — gigabytes of NASA elevation data, a custom mesh pipeline, PETG on hardboard, projection-mapped.</li><li>• <b>Expert Witness, EFF</b> — testified in U.S. District Court (SDNY) as a technical expert in the Electronic Frontier Foundation's <b>Privacy Act</b> <u>litigation over access to federal personnel systems</u>.</li><li>• <b>Network-intrusion investigation</b> — custom Wireshark <u>dissectors</u> for Ubiquiti and HikVision NVR protocols, written to investigate a network intrusion.</li><li>• <b>Guest lecturer, Vanderbilt Policy Accelerator</b> — on resiliency and privacy engineering.</li></ul> |
| 2024 – 25  | <div>AI &amp; Cybersecurity Advisor / to the Federal CIO</div> <div>WHITE HOUSE / OMB</div> <p>Cybersecurity, AI, and modernization advisor to the Federal CIO and OMB CIO — doing everything from engineering to policy analysis.</p> <ul style="list-style-type: none"><li>• Led a cross-functional <b>5-person sprint team</b> to assess the agency's IT modernization needs, and drafted a <b>~\$100M</b> modernization proposal for the CIO.</li><li>• Designed and led the agency's first <b>AI Community of Practice</b>, and began building the agency's AI program.</li><li>• Designed and built a AI platform for on-premise models on <b>NVIDIA A100</b> hardware and Kubernetes, enabling teams to begin building prototypes.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

2021 — 23

## SRE & Infrastructure Manager / ~12 engineers

REBELLION DEFENSE

Ran reliability & infrastructure for an AI-powered defense startup — AWS, Kubernetes, novel infra. Roughly half the job stayed hands-on as an IC.

2019 — 21

## Deputy Chief Information Officer / Term SES

US OPM

- Under the CIO, led a **~700-person** organization with a **\$100M** annual budget, and introduced the agency's first cloud-based products.
- Built the agency's first **Digital Services team** — from writing the position descriptions through recruiting and hiring.
- Conducted multiple briefings to **members of Congress**, including in classified settings.
- **Correspondence-system rescue** — OPM's correspondence-tracking system died and no one knew how it worked; reverse-engineered it and shipped a **Go** stop-gap so staff could reach and archive their records.
- **Single point of failure** — OPM's two mainframes, meant to fail over for each other, had drifted into co-located shared capacity with no real redundancy. Spotted it during the OPM/DCSA disentanglement and led the remediation — four right-sized chassis split across separate data centers.
- **Virtual call center** — a call-center failure was pinned on a multi-year fix; a working **Twilio reproduction**, built in **~3 hours**, proved it wasn't.

2014 — 19

## Director of Engineering / SRE & Security · founding member

US DIGITAL SERVICE

**Founding member** of the U.S. Digital Service (2014) — from the HealthCare.gov stabilization through rapid-response & modernization across HHS/CMS, GSA, State and DOJ. As Director of Engineering, **mentored ~50 engineers and directly supervised 25**, and co-designed the engineering hiring pipeline.

- Regularly briefed and advised **agency heads and White House cabinet officials**.
- **Rapid-response sprints** — led short-duration engineering teams dropped into failing federal programs: triage, stabilize, and hand back a system that works.
- **Auto-fixer for vulns** — a static analyzer that generated fixes for **thousands** of security vulnerabilities in a legacy government system.
- **College Scorecard** — UX research, design, and the initial prototype of the application and data pipeline behind the U.S. Dept. of Education's open college-outcomes data.

2007 — 14

## Site Reliability Engineer / logs infrastructure

GOOGLE

Full-stack ownership for O(1M) CPU cores and exabytes of data — cryptographic data-integrity proofs, petabyte cross-cluster migrations, and a role in Google's response to state-sponsored intrusions.

1995 — 07

## Technical Architect & Engineer

AT&T

Technical lead for att.com; introduced the DevOps model to production operations.

1998

## Computer Engineering / Texas A&M University

EDUCATION